

BEST PRACTICES FOR SEALING HVAC SYSTEMS

Energy Star programs are requiring more expertise from HVAC crews. Here are some helpful training hints.

by **BRUCE MANCLARK**

Recently I wrote a guide for the Northwest Energy Star New Homes program. The guide is designed to help instructors address the training needs of HVAC installation crews who have little or no experience in duct leakage testing. The goal of this training is to give HVAC contractors the information they need to seal new HVAC systems effectively. Key to achieving this goal is the trainer's ability to mesh the techniques of creating an Energy Star sealed air duct system with the manufacturer's new-construction installation procedures.

In order to achieve this goal, it is necessary to train more than the initial installation crew. In most large HVAC shops, different crews and individuals perform different tasks related to installing an airtight duct system. Training may involve spending two hours at an actual job site, instructing crews on the best ways to apply mastic. But training should also involve spending time with the new installation manager, the sales staff, the purchasing manager, and the top-out crew. It is essential to remember that you are training the whole company, not just the install crew.

Asking the Right Questions

As a trainer, you should understand how each shop sequences its HVAC jobs. The first key information you must gather is who does what and when during the installation process. Learn



Training in the use of tools such as the Duct Blaster shown above has helped installation crews meet Northwest Energy Star New Home standards for duct tightness.

how the jobs are sequenced by talking with job managers, schedulers, and key crew members. Once you have this information, you can decide how best to integrate duct sealing into the shop's installation process. You will need to get the answers to two sets of questions:

1. How many site trips are being made on each HVAC installation job? Be aware that there might be separate trips for
 - crawlspace work;
 - attic work;
 - air handler setting;
 - fire-up; and
 - grille setting.

2. Who are the key players on the HVAC crew?
 - Who does what?
 - When do they do it?

Depending upon the installation sequence and the individual roles and responsibilities of crew members, sealing the entire duct system may take several visits to the job site by several different people. Make sure that every key member of the HVAC crew is able to attend the training; otherwise, it will be difficult to train the crew properly.

Get complete answers to all the questions. These answers will help you to engage the techs and to suggest to the crews how best to integrate duct sealing into their usual installation sequence. Remember that there is no "one size fits all" process for duct sealing. The process you develop for each company and crew should be customized to the size of the company and the individual roles and responsibilities of the crew members.

Training Tips

Make sure that you speak the crew's language. The phrase "CFM at 50 Pascals" has confused more HVAC crew members than the International Mechanical Code. They might as well have heard, "RPM at 50 rascals." While crews trained in the use of a Duct Blaster or blower door will recognize it as a measurement of volumetric air flow at 50 Pa of pressure difference, crews that have only sealed ducts without experience

testing for tightness will not know the language. Not very many crews know that a Pascal is a Newton per meter squared, let alone have any intuitive understanding of what this means.

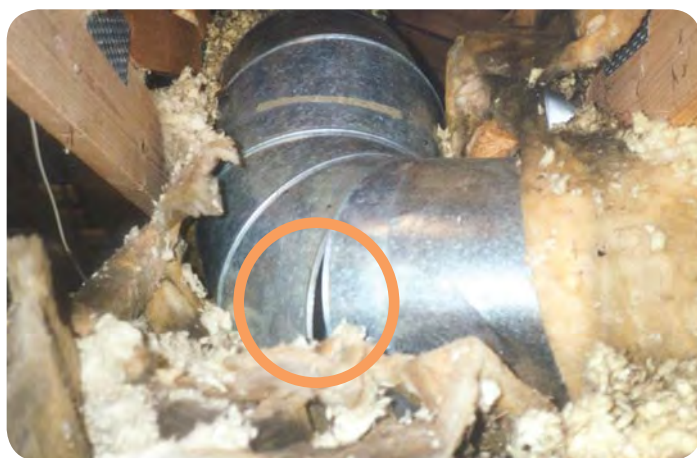
It is better to describe leakage in terms of allowed leakage area in square inches. While this will not give a precise measurement, an estimate of area leakage can be obtained by dividing CFM at 50 Pa (CFM_{50}) by 10. Using the allowed leakage area in the Northwest Energy Star program of 6% of the floor area at

thing somebody is making me do.” This is a pretty universal response—one that may cause some uneasiness for you as the trainer. However, it gives you a prime opportunity to serve as the solution, rather than the problem.

Crews installing in new homes are expected to get duct systems installed ASAP. A work slowdown caused by duct sealing could delay the entire construction process. The job then goes over cost, and the boss is not happy. Typically, crews are expected to meet the addi-

chemical smoke pencil or a bubble solution to make leaks (both in and out of the system) more visible. You can use the same duct system model to demonstrate how to apply duct mastic correctly (especially on gored elbows) and then retest the system for leaks.

When crews do their own duct leakage tests in training, it helps them create their own internal feedback loops. The more real the training is, the better it is at establishing these loops.



The crack in this duct elbow gore has compromised the HVAC system in this house.



The site of leakage can be identified by the dirty, discolored insulation.

50 Pa, the allowed area of holes would be 6 square inches per 1,000 square feet of home. Similarly, a 2,000 ft² home would have a maximum allowed leakage of 12 square inches.

Using square inches rather than flow to describe duct leakage gives you the opportunity to describe the leakage of the individual duct components in an aggregate way that is easy to understand. To convince crews that there is that much leakage in a duct system, use a little math. We've measured 5 CFM_{50} at unsealed elbows. Dividing 5 CFM_{50} by 10 gives 1/2 square inch of leakage area per elbow. They will see how it all adds up.

Give Them Respect

You may discover that the crews you work with view Energy Star duct sealing and testing as “just another darn

tional requirements of Energy Star in the same amount of time that it would take them to install a conventional duct system. You need to teach them how to work efficiently with the necessary tools and materials in order to get the job done quickly and correctly.

Repetition is the key to getting adults to retain knowledge. The old adage Tell them what you are going to tell them, tell them, and then tell them what you just told them works well in this environment. And while you're repeating your message, try to keep the training enjoyable. The more props you use, the easier it will be to keep the crew's attention. Pictures are good, and a live model is even better. Because seeing is believing, set up a few tests to engage your audience.

For instance, you can set up a bench-top duct system to illustrate how much various duct components leak. Use a

The 5 Ps of Airtight Duct Installation

Achieving success at installing an airtight duct system depends on the 5 Ps:

- Plan
- Purchase
- Prioritize
- Plug
- Preseal

Plan

Planning means scheduling who does what, and when they do it. Asking the questions posed in the introduction will help you assign the tasks necessary to conduct the various tests. Here is a sample list of tasks for a typical shop:

1. Preseal the boots, elbows, and Ys.
2. Install and seal crawlspace ducts (if weather prohibits duct sealing, assign task to other crew).

3. Install and seal air handler and attic ducts.
4. Seal boots to subfloor.
5. Seal ceiling registers to Sheetrock.
6. Install grilles and registers.
7. Tape and plug holes in air handler and air handler connections.

Purchase

Purchase sheet-metal parts that require little or no sealing. Nearly all distributors stock HVAC fittings that require minimal sealing. Airtight takeoffs (ATOs)—an alternative to gored elbows—are a good example. These takeoffs have no gores; they come with a sticky foam-backed flange and help increase air flow to individual duct runs. If applied correctly, ATOs require only a quick brush of mastic. Other examples of HVAC fittings that require little or no sealing include

- airtight filter cabinets;
- air handlers with gasket sealing;
- electrical and piping penetrations;
- presealed fittings supplied by the manufacturer; and
- Speedi-Boots or similar products that create an airtight seal between the Sheetrock and the ceiling boot or box.

Prioritize

It is easy to say that in new construction every joint, crack, hole, and seam should be sealed, but the reality is that there is never enough time or labor budget to do so. By prioritizing the sections of the duct system that leak the most (especially under operating conditions), crews can focus on these areas first. Most duct systems in the Northwest are flex duct systems, and properly installed flex ducts do not leak. Proper installa-

tion procedure consists of a tension tie placed according to manufacturer's instructions, using a tensioning tool, around the interior and exterior liners.

In addition, various code enforcement agencies require the use of mastic or tape around the flex-to-takeoff connection. At best this does little to reduce leakage. Requiring crews to smear mastic around a takeoff or the inside liner of flex duct makes the job harder,



Poor duct sealing is only one of the many problems that can be effectively corrected with better HVAC-training programs.

slower, and messier. If the durability of the takeoff-to-flex duct connection is in question, beaded start collars or screws placed under (not through) the tension tie will do a better job.

The bigger flexible elbows are, the more they leak. A 6- or 8-inch elbow leaks approximately 5 CFM at 50 Pa. When you consider that the typical system has a total of 10 to 20 elbows, it's easy to see that the leaks can really add up. What makes matters worse is that elbows are hard to tighten. To thoroughly seal these components, mastic must be worked into the gores. Flexible elbows are usually placed in the 90° position and can be presealed quite easily.

Although boots don't usually leak much, they are easy to seal, and sealing them may make the difference between passing or failing the test. They are also easy to preseal. Make sure that crews

also check the boot-to-floor connection. The gap between the floor and the subfloor is not a leak in the duct system. However, under certain test procedures, the Duct Blaster may read this gap.

When a test is conducted with the registers in place and taped off, the air from the Duct Blaster may be trapped by the tape covering the register and leak out of the gap between the subfloor and the boot, or the gap between the ceiling boot or the B-box and the Sheetrock. This will not happen if foam blocks are used, because the air is stopped at the boundary of the foam block.

If your program uses tape to seal off the registers and grilles, it is imperative to seal the gap between the boot and the ceiling, the boot and the floor, or the boot and the wall. Unfortunately, this cannot be done at the point of installation. It must be done by the top-out crew after the Sheetrock is installed. Using a Speedie Boot, crews can make this connection tight at the time of installation. This specially

designed boot has a gasket that seals against the backside of the Sheetrock or the subfloor.

Another method of sealing a B-box to the Sheetrock is to apply ³/₈-inch-thick closed cell foam to the flange of the B-box. This allows the Sheetrock to be pressed against the foam to create an airtight barrier.

The use of building cavities to convey air is not recommended in the Energy Star program, and for good reason. Cavities leak a lot. It is usually easier to install hard ducts than it is to seal building cavities. If the crew cannot seal the cavity, lining it with Thermo Ply (or a similar product) produces good results. When a building cavity is lined with thermo pan, use mastic to ensure tightness.

In many cases, when building cavities are used as ductwork, often the electri-

BRUCE MANCLARK

cal and plumbing contractors will penetrate the cavities with electric wiring and water distribution pipes after the ducts are sealed and tested. Checking for and sealing these penetrations is the responsibility of the top-out crew.

It is extremely important to seal the takeoffs at the supply plenum. They can leak a lot, because they are difficult to seal and are often under high pressure when the system is running. Dove-tailed take offs may require multiple attempts to seal because of the gaps that dovetails often form. An airtight takeoff (ATO), which was discussed earlier, makes the job easier—but the connection still needs to be sealed.

The Energy Star program allows the furnace-to-plenum connection to be sealed with a metal UL181-approved tape. Make sure that all four sides of the furnace to plenum connection are sealed. The backside is often ignored because it is difficult to reach.

The biggest leak in the system may be in the air handler cabinets.

Air handlers typically leak 20–100 CFM at 50 Pa. The Northwest Energy Star program allows the cabinet to be sealed with UL181-approved metal tape. All the penetrations for gas and condensate lines in the air handler and A/C cabinets should be sealed. In systems where air conditioning is installed, it is highly recommended that condensate plumbing penetrations be sealed with a waterproof material such as cork tape. Taping the cabinet doors is allowed under the Energy Star program. Never tape over any electrical breakers.

If the builder is using toe kicks (that is, a grille below the face of the kitchen/bathroom cabinet), it is imperative that the duct entrance point be sealed where it penetrates the subfloor. Installed sheet-metal hoods that direct the airflow into the room will help seal toe kicks, as well as increase the amount of air entering the room. As a testing note, the toe kicks should always be sealed at the entry point. Otherwise, the Duct

Blaster may be testing the integrity of the wall behind the cabinet.

Finally, the use of high-quality filter slots is not an industry standard in new construction. Obtain the best results by sealing the filter slot on the three sides of the access door, and avoid guillotine-type filter slots.

Plug and Preseal

Plug; don't paint. Crews often apply mastic to all the joints they can, and still

plenum that is installed in the crawlspace. This includes installing the takeoffs. This can be a big benefit, especially in wet or cold weather.

Water-based mastics, by nature, cannot be installed in the rain. This unfortunate attribute mainly affects ducts installed in the crawlspace before the subfloor is in place. In cases where the air handler is located in the garage, the stub plenum extends into the crawlspace. The takeoffs need to be sealed well. In many cases, mastic may have to be applied when the subfloor is in place and conditions are dry.

Feedback and Follow-Up

Once crews have been trained on the proper duct sealing protocols and have implemented what they have learned, they need immediate feedback as to whether the jobs they do pass or fail. If the job fails, tell them why. If possible, test a few jobs while the crew is finishing up.

In the absence of crew testing, good, timely and helpful feedback will be the single biggest factor in helping crews improve their work over time.

Using the type of training outlined in this article has resulted in installation crews that, in a short time, can meet the duct tightness standards of the Northwest Energy Star New Homes Program. In the Northwest, we are now seeing HVAC shops that have been delivering high performance HVAC systems for over five years develop the ability to train their own employees. This is a very welcome development, and is proof that training programs that increase the chances of early success can help HVAC contractors make the changes in installation practices that insure efficient heating and cooling systems.



Bruce Manclark is a member of the Northwest Energy Star New Homes Team, specializing in HVAC training.



BRUCE MANCLARK

Effective training programs can ensure better results, such as this well-sealed elbow.

their duct sealing job fails. This is because they did not apply enough mastic. Mastic is not paint; it must be applied thick enough to plug the leaks. During training, equate the proper use of mastic with the use of pipe dope in water or gas piping. If you don't put enough on the joint, it will leak. Most mastic manufacturers state that mastic should be $\frac{1}{16}$ inch thick, about the thickness of a nickel. Applying mastic at the correct thickness will also help keep the seal from cracking over time.

Presealing is the process of applying mastic to duct parts and then allowing the mastic to seal before installation begins. Presealing can take place at the shop, where work conditions are warm and dry. I highly recommend presealing connections like boots and Y-joints. Applying mastic before the installation process makes the job cleaner, faster, and less messy.

When conditions allow, many companies preinsulate and preseal the supply